

D_{s1} normalized-current $AA/AB/BB$ two-point QCD sum rule

Calculation note

July 29, 2026

Abstract

This note replaces the former external- f_1 /overlap normalization of the $D_{s1}(2460)$ and $D_{s1}(2536)$ currents. One normalized-current $AA/AB/BB$ two-point QCD sum-rule matrix determines the mixing angle and both physical residues. The calculation is implemented independently in Python and Mathematica. At the stated $LO+d=3+d=5$ truncation the central Mathematica and Python results agree below 10^{-7} .

1 Scope and the local-condensate issue

This is an ordinary two-point SVZ sum rule. Therefore the local vacuum condensates $\langle \bar{s}s \rangle$ and $\langle \bar{s}g_s\sigma \cdot Gs \rangle$ are part of its OPE. They are not inserted into the external-photon transition LCSR, which follows the Rohrwild organization in terms of non-local photon matrix elements and photon distribution amplitudes. The two statements concern different correlators and are fully compatible.

The present common matrix truncation contains

- exact- m_s leading-order perturbative spectral densities;
- the local strange condensate through $O(m_s^2)$;
- the local dimension-five mixed condensate.

The gluon condensate and perturbative α_s corrections are not included. Thus “complete” below means a complete 2×2 current matrix at this explicit common truncation, not an all-dimensional or all-orders OPE.

2 Currents, correlator, and rotation

The normalized basis currents are

$$J_A^\mu = \bar{s}\gamma^\mu\gamma_5 c, \quad J_B^\mu = \frac{i}{m_c + m_s} \bar{s}\sigma^{\mu\nu} p'_\nu \gamma_5 c, \quad (1)$$

where p' is the initial-state momentum. The physical currents are

$$J_1^\mu = \sin\theta J_A^\mu + \cos\theta J_B^\mu, \quad (2)$$

$$J_2^\mu = \cos\theta J_A^\mu - \sin\theta J_B^\mu. \quad (3)$$

For $i, j \in \{A, B\}$,

$$\Pi_{ij}^{\mu\nu}(p') = i \int d^4x e^{ip' \cdot x} \langle 0 | T \{ J_i^\mu(x) J_j^{\nu\dagger}(0) \} | 0 \rangle = \left(-g^{\mu\nu} + \frac{p'^\mu p'^\nu}{p'^2} \right) \Pi_{ij}(p'^2) + \dots \quad (4)$$

Writing

$$R(\theta) = \begin{pmatrix} \sin \theta & \cos \theta \\ \cos \theta & -\sin \theta \end{pmatrix}, \quad \Pi^{\text{phys}} = R \Pi^{\text{basis}} R^T, \quad (5)$$

gives

$$\Pi_{11} = \sin^2 \theta \Pi_{AA} + \cos^2 \theta \Pi_{BB} + 2 \sin \theta \cos \theta \Pi_{AB}, \quad (6)$$

$$\Pi_{22} = \cos^2 \theta \Pi_{AA} + \sin^2 \theta \Pi_{BB} - 2 \sin \theta \cos \theta \Pi_{AB}, \quad (7)$$

$$\Pi_{12} = \sin \theta \cos \theta (\Pi_{AA} - \Pi_{BB}) + (\cos^2 \theta - \sin^2 \theta) \Pi_{AB}. \quad (8)$$

Equation (6) follows directly from the stated definition of J_1 . The early draft had the AA and BB coefficients interchanged in this equation; that version is incorrect.

3 Perturbative spectral matrix

Let $d = m_c + m_s$ and

$$\lambda(s) = [s - (m_c + m_s)^2][s - (m_c - m_s)^2]. \quad (9)$$

The exact-mass LO densities are

$$\rho_{AA}(s) = \frac{\sqrt{\lambda}}{8\pi^2} \left[2 - \frac{m_c^2 + m_s^2 + 6m_c m_s}{s} - \frac{(m_c^2 - m_s^2)^2}{s^2} \right], \quad (10)$$

$$\rho_{AB}(s) = \rho_{BA}(s) = \frac{\sqrt{\lambda}}{8\pi^2} \frac{3(m_s - m_c)(d^2 - s)}{ds}, \quad (11)$$

$$\rho_{BB}(s) = \frac{\sqrt{\lambda}}{8\pi^2} \frac{-(d^2 - s)(2m_s^2 - 4m_c m_s + 2m_c^2 + s)}{d^2 s}. \quad (12)$$

The AA density is algebraically identical to the Colangelo axial-current two-point density. The equality and $AB = BA$ are exact FeynCalc regression checks, not numerical observations.

4 Local Borel matrix

With $C_3 = \langle \bar{s}s \rangle$ and $C_5 = \langle \bar{s}g_s \sigma \cdot Gs \rangle$, the continuum-subtracted Borel matrix is

$$\hat{\Pi}(M^2, s_0) = \int_{d^2}^{s_0} ds e^{-s/M^2} \rho(s) + e^{-m_c^2/M^2} [C_3(\mathcal{Q}_0 + \mathcal{Q}_1 + \mathcal{Q}_2) + C_5 \mathcal{Q}_5]. \quad (13)$$

The local matrices are

$$\mathcal{Q}_0 = \begin{pmatrix} m_c & m_c^2/d \\ m_c^2/d & m_c^3/d^2 \end{pmatrix}, \quad (14)$$

$$\mathcal{Q}_1 = \begin{pmatrix} -\frac{m_s m_c^2}{2M^2} & \frac{m_s m_c (1 - m_c^2/M^2)}{2d} \\ \frac{m_s m_c (1 - m_c^2/M^2)}{2d} & \frac{m_s m_c^2 (1 - m_c^2/(2M^2))}{d^2} \end{pmatrix}, \quad (15)$$

$$\mathcal{Q}_2 = \begin{pmatrix} \frac{m_s^2 m_c^3}{2M^4} & \frac{m_s^2 (m_c^4/M^4 - m_c^2/M^2 - 1)}{d^2} \\ \frac{m_s^2 (m_c^4/M^4 - m_c^2/M^2 - 1)}{2d} & \frac{m_s^2 (m_c^5/(2M^4) - m_c^3/M^2)}{d^2} \end{pmatrix}, \quad (16)$$

$$\mathcal{Q}_5 = \begin{pmatrix} -\frac{m_c^3}{4M^4} & -\frac{m_c^4/M^4 - m_c^2/M^2 - 1}{4d} \\ -\frac{m_c^4/M^4 - m_c^2/M^2 - 1}{4d} & -\frac{m_c^5/(4M^4) - m_c^3/(2M^2)}{d^2} \end{pmatrix}. \quad (17)$$

The complete AA local entry reduces exactly to

$$e^{-m_c^2/M^2} \left\{ C_3 \left[m_c - \frac{m_c^2 m_s}{2M^2} + \frac{m_c^3 m_s^2}{2M^4} \right] - C_5 \frac{m_c^3}{4M^4} \right\}, \quad (18)$$

which is the corresponding Colangelo expression.

5 Fixed angle, thresholds, and residues

The nominal angle is taken from our dedicated mixing-angle study,

$$\theta_{D_s} = 26.6 \pm 0.6^\circ. \quad (19)$$

The angle obtained from $\hat{\Pi}_{12} = 0$ in the present $LO+d=3+d=5$ matrix is retained only as a truncation diagnostic. We fit a separate threshold $s_0^{(i)}$ to each measured pole and use

$$m_{i,\text{eff}}^2 = -\frac{d}{d(1/M^2)} \ln \hat{\Pi}_{ii}(M^2, s_0^{(i)}; \theta), \quad (20)$$

$$f_i^2 m_i^2 e^{-m_i^2/M^2} = \hat{\Pi}_{ii}(M^2, s_0^{(i)}; \theta), \quad i = 1, 2. \quad (21)$$

In 2000 input samples, 1714 pass all diagnostics. The result is

$$\theta_{D_s} = 26.595[25.996, 27.199]^\circ, \quad (22)$$

$$\theta_{\text{matrix}} = 30.632[29.942, 31.286]^\circ, \quad (23)$$

$$f_1 = 0.4046[0.3799, 0.4291] \text{ GeV}, \quad f_2 = 0.1677[0.1594, 0.1756] \text{ GeV}. \quad (24)$$

The normalized off-diagonal residual after projecting at the external angle is $0.183[0.142, 0.225]$. It is an OPE-truncation diagnostic, not an additional statistical angle error. Neither residue is externally anchored.

At $M^2 = 2.35 \text{ GeV}^2$, the fixed-angle Python regression point is

$$\theta = 26.6^\circ, \quad \theta_{\text{matrix}} = 30.58596159^\circ, \quad s_0^{(1)} = 7.925 \text{ GeV}^2, \quad s_0^{(2)} = 9.975 \text{ GeV}^2, \quad (25)$$

$$f_1 = 0.4005893052 \text{ GeV}, \quad f_2 = 0.1666703674 \text{ GeV}. \quad (26)$$

The independent Mathematica regression gives

$$f_1 = 0.4005892785 \text{ GeV}, \quad f_2 = 0.1666703614 \text{ GeV}. \quad (27)$$

The differences from Python are $-2.67 \times 10^{-8} \text{ GeV}$ and $-6.02 \times 10^{-9} \text{ GeV}$, respectively; the matrix-angle and normalized Π_{12} diagnostics agree to better than 10^{-7} .

6 Propagation, interference, and final charm widths

The two transition invariants give the physical couplings. Their continuum factors are term-specific: twist 2 uses $\Delta E_Q = e^{-m_Q^2/M^2} - e^{-s_0/M^2}$; twist 3, twist 4, and the gluonic three-particle term use $E_Q = e^{-m_Q^2/M^2}$; and the separate gauge-completion electromagnetic term uses ΔE_Q :

$$G_{2460} = \frac{e^{(m_{2460}^2 + m_{D_s}^2)/(2M^2)} (m_c + m_s)}{m_{2460} f_1 m_{D_s}^2 f_{D_s}} \left(\sin \theta \hat{T}_A + \cos \theta \hat{T}_B \right), \quad (28)$$

$$G_{2536} = \frac{e^{(m_{2536}^2 + m_{D_s}^2)/(2M^2)} (m_c + m_s)}{m_{2536} f_2 m_{D_s}^2 f_{D_s}} \left(\cos \theta \hat{T}_A - \sin \theta \hat{T}_B \right). \quad (29)$$

For the preferred lattice photon normalization, the final-window Monte Carlo gives

$$G_{2460} = -0.4082[-0.4551, -0.3654] \text{ GeV}^{-1}, \quad \Gamma_{2460} = 34.96[28.06, 43.53] \text{ keV}, \quad (30)$$

$$G_{2536} = +0.05184[+0.00244, +0.1067] \text{ GeV}^{-1}, \quad \Gamma_{2536} = 0.988[0.092, 3.555] \text{ keV}. \quad (31)$$

The low-state pieces are constructive, whereas the high-state pieces are destructive throughout the physical first quadrant. At the midpoint of the selected transition window the benchmark widths are

Angle input	θ	Γ_{2460} (keV)	Γ_{2536} (keV)
previous study	26.6°	35.57	0.664
truncated-matrix diagnostic	30.61°	35.06	4.410
HQET benchmark	35.3°	34.23	12.32

The high-state zero is near 23.96° at this representative point. The interference mechanism does not change; the relative weights move the result toward or away from the cancellation.

7 Projection onto Wang’s pure-AA channel

Wang’s arXiv v4 result is $f_{D_{s1}}^{(JA)} = 0.345 \pm 0.017 \text{ GeV}$; the earlier $0.245 \pm 0.017 \text{ GeV}$ entry was explicitly corrected. Reproducing this one-pole number is neither necessary nor sufficient to validate the mixed residues. The inverse current rotation is

$$J_A = \sin \theta J_1 + \cos \theta J_2, \quad (32)$$

so our physical poles project back to

$$\hat{\Pi}_{AA}^{(2\text{pole})} = \sin^2 \theta f_1^2 M_1^2 e^{-M_1^2/M^2} + \cos^2 \theta f_2^2 M_2^2 e^{-M_2^2/M^2}. \quad (33)$$

The corresponding one-pole-equivalent diagnostic at the lower mass is

$$f_{A,\text{eq}}^2(M^2) = \frac{e^{M_1^2/M^2}}{M_1^2} \hat{\Pi}_{AA}^{(2\text{pole})} \quad (34)$$

and gives $0.2303[0.2194, 0.2418] \text{ GeV}$. Separately, in Wang’s Borel and threshold window our direct pure-AA LO+ $d = 3+d = 5$ OPE gives

$$f_A^{\text{ours}} = 0.2686[0.2607, 0.2768] \text{ GeV}, \quad (35)$$

which does not reproduce Wang’s absolute normalization. These three numbers use different pole models and OPE truncations. They are useful normalization benchmarks, not a proof or disproof of the mixed-current calculation. Neither f_1 nor f_2 should be equated to Wang’s number. The individual projected overlaps are

$$\sin \theta f_1 = 0.1812[0.1692, 0.1930] \text{ GeV}, \quad (36)$$

$$\cos \theta f_2 = 0.1498[0.1424, 0.1573] \text{ GeV}, \quad (37)$$

These are projection diagnostics only.

8 Direct bottom-strange continuation

The same spectral and local matrices apply after $m_c \rightarrow m_b$. Projecting at the external $\theta_{B_s} = 38.5 \pm 0.1^\circ$ angle and fitting separate thresholds to the model lower pole and the observed high pole gives, at $M^2 = 7.00 \text{ GeV}^2$,

$$s_0^{(1)} = 44.35 \text{ GeV}^2, \quad s_0^{(2)} = 43.025 \text{ GeV}^2, \quad (38)$$

and

$$f_1^{(B_s)} = 0.5381107 \text{ GeV}, \quad f_2^{(B_s)} = 0.0888787 \text{ GeV}. \quad (39)$$

The pole fractions are 0.805 and 0.556 and the dimension-five fractions are 0.0029 and 0.0348. The matrix-diagonalization diagnostic is 39.50° and the normalized off-diagonal residual at the external angle is 0.104.

Scanning $M^2 = 6.50\text{--}7.30 \text{ GeV}^2$ and varying the OPE inputs gives

$$f_1^{(B_s)} = 0.5358[0.5096, 0.5586] \text{ GeV}, \quad f_2^{(B_s)} = 0.08896[0.08496, 0.09278] \text{ GeV}. \quad (40)$$

The preferred radiative widths after propagating these direct residues are

$$\Gamma[B_{s1}(5750) \rightarrow B_s \gamma] = 11.34[9.144, 14.06] \text{ keV}, \quad (41)$$

$$\Gamma[B_{s1}(5830) \rightarrow B_s \gamma] = 8.702[6.474, 11.23] \text{ keV}. \quad (42)$$

The lower pole is still a model-state diagnostic, but no external Pullin–Zwicky decay constant or overlap closure is used.

9 Executable record

- Python matrix and Monte Carlo: `../scripts/twopoint_ds1_matrix_sumrule.py`.
- Spectral trace audit: `../scripts/twopoint_dirac_spectral_audit.wl`.
- Local trace audit: `../scripts/twopoint_local_condensate_audit.wl`.
- Numerical Mathematica regression: `../scripts/mathematica_twopoint_ds1_matrix_check.wl`.
- Direct bottom Python matrix and Monte Carlo: `../scripts/twopoint_bs1_matrix_sumrule.py`.
- Direct bottom Mathematica regression: `../scripts/mathematica_twopoint_bs1_matrix_check.wl`.
- Cell-by-cell notebook: `../notebooks/Ds1_AA_AB_BB_two_point_sumrule.nb`.

The former $f_1 = 0.345 \text{ GeV}$, $f_2 \simeq 0.38 \text{ GeV}$ overlap model is preserved only as a legacy comparison and is not the current paper normalization.